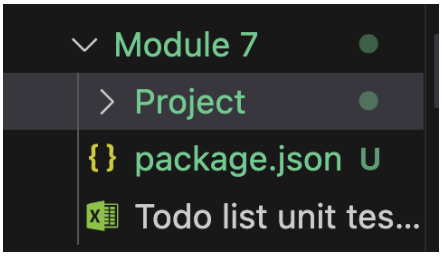
Param Dholakia

Module 7-Unit Testing

Tests done:

1. Add Todo form should be created when addTodo() is called.
2. Cancel button should remove the Add Todo form.
3. Clicking Save with valid User ID and Title should store a todo in localStorage.
4. Clicking Save with empty inputs should NOT store anything in localStorage.

Folder Structure:



npm init -y

```
param@Params-MacBook-Air Module 7 % npm init -y
Wrote to /Users/param/Desktop/gitlab/param-h-dholakia/09. qa-testing/Module 7/package.json:

{
  "name": "module-7",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}

param@Params-MacBook-Air Module 7 %
```

Install Jest

npm install --save-dev jest

```
param@Params-MacBook-Air Module 7 % npm install --save-dev jest
npm warn deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do not use it. Check out lru-cache if you want
a good and tested way to coalesce async requests by a key value, which is much more comprehensive and powerful.
npm warn deprecated glob@7.2.3: Old versions of glob are not supported, and contain widely publicized security vulnerabilities, whi
ch have been fixed in the current version. Please update. Support for old versions may be purchased (at exorbitant rates) by contac
ting i@izs.me
npm warn deprecated glob@10.5.0: Old versions of glob are not supported, and contain widely publicized security vulnerabilities, wh
ich have been fixed in the current version. Please update. Support for old versions may be purchased (at exorbitant rates) by conta
cting i@izs.me

added 298 packages, and audited 299 packages in 18s

44 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
param@Params-MacBook-Air Module 7 %
```

For DOM Testing

npm install --save-dev jest-environment-jsdom

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```
param@Params-MacBook-Air Module 7 % npm install --save-dev jest-environment-jsdom
npm warn deprecated whatwg-encoding@3.1.1: Use @exodus/bytes instead for a more spec-conformant and faster implementation

added 42 packages, and audited 341 packages in 5s

51 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
param@Params-MacBook-Air Module 7 %
```

<u>As my project uses Fetch api</u>

npm install --save-dev jest-fetch-mock

```
param@Params-MacBook-Air Module 7 % npm install --save-dev jest-fetch-mock

added 7 packages, and audited 348 packages in 3s

51 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
param@Params-MacBook-Air Module 7 %
```

Update package.json

Old file:

```
{} package.json U ×
param-h-dholakia > 09. qa-testing > Module 7 > {} package.json > {} devDepen
1  {
2    "name": "module-7",
3    "version": "1.0.0",
4    "description": "",
5    "main": "index.js",
       ▷Debug
6    "scripts": {
7      "test": "echo \"Error: no test specified\" && exit 1"
8    },
9    "keywords": [],
10   "author": "",
11   "license": "ISC",
12   "type": "commonjs",
13   "devDependencies": {
14     "jest": "^30.2.0",
15     "jest-environment-jsdom": "^30.2.0",
16     "jest-fetch-mock": "^3.0.3"
17   }
18 }
19
```

New File:

```
{
  "name": "module-7",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  ▷Debug
  "scripts": {
    "test": "jest"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs",
  "jest": {
    "testEnvironment": "jsdom"
  },
  "devDependencies": {
    "jest": "^30.2.0",
    "jest-environment-jsdom": "^30.2.0",
    "jest-fetch-mock": "^3.0.3"
  }
}
```

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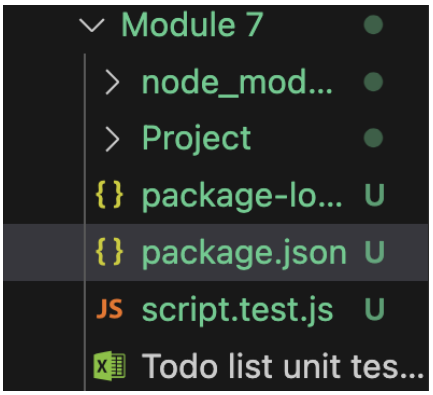
Create a new file:

script.test.js



Run :

npm test

```
param@Params-MacBook-Air Module 7 % npm test

> module-7@1.0.0 test
> jest

 PASS  ./script.test.js
  Add Todo Module
    ✓ Add button should create input form (8 ms)
    ✓ Cancel button should remove form (2 ms)
    ✓ Save should store todo in localStorage (1 ms)
    ✓ Should not save if inputs are empty (1 ms)

Test Suites: 1 passed, 1 total
Tests:       4 passed, 4 total
Snapshots:   0 total
Time:        0.329 s, estimated 1 s
Ran all test suites.
param@Params-MacBook-Air Module 7 %
```

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TEST SNIPPETS

TEST 1:

```
// Test 1
 test("Add button should create input form", () => {
  const main = document.querySelector(".maindisplay");
  const bottom = document.createElement("div");
  main.appendChild(bottom);

  addTodo(main, bottom);

  const form = document.querySelector(".add-todo");
  expect(form).not.toBeNull();
 });
```

TEST 2:

```
// Test 2
 test("Cancel button should remove form", () => {
  const main = document.querySelector(".maindisplay");
  const bottom = document.createElement("div");
  main.appendChild(bottom);

  addTodo(main, bottom);

  const buttons = document.querySelectorAll(".add-todo button");
  const cancelBtn = [...buttons].find(btn => btn.textContent === "Cancel");

  expect(cancelBtn).toBeDefined();   // Safety check
  cancelBtn.click();

  const form = document.querySelector(".add-todo");
  expect(form).toBeNull();
 });
```

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TEST 3:

```
// Test 3
 test("Save should store todo in localStorage", () => {
   const main = document.querySelector(".maindisplay");
   const bottom = document.createElement("div");
   main.appendChild(bottom);

   addTodo(main, bottom);

   const inputs = document.querySelectorAll(".add-todo input");
   inputs[0].value = "1";
   inputs[1].value = "Test Title";

   const buttons = document.querySelectorAll(".add-todo button");
   const saveBtn = [...buttons].find(btn => btn.textContent === "Save");

   saveBtn.click();

   expect(localStorage.length).toBe(1);
 });
```

TEST 4:

```
// Test 4
 test("Should not save if inputs are empty", () => {
   const main = document.querySelector(".maindisplay");
   const bottom = document.createElement("div");
   main.appendChild(bottom);

   addTodo(main, bottom);

   const buttons = document.querySelectorAll(".add-todo button");
   const saveBtn = [...buttons].find(btn => btn.textContent === "Save");

   saveBtn.click();

   expect(localStorage.length).toBe(0);
 });
```